

Foundations of Multi-armed bandit

[Multi-armed bandit]

1.0 Content Block

Collaborative bandit Approaches using multiple bandits that cooperate sharing knowledge in order to better optimize their performance started in 2013 with "A Gang of Bandits", an algorithm relying on a similarity graph between the different bandit problems to share knowledge. The need of a similarity graph was removed in 2014 by the work on the CLUB algorithm. Following this work, several other researchers created algorithms to learn multiple models at the same time under bandit feedback. For example, COFIBA was introduced by Li and Karatzoglou and Gentile (SIGIR 2016), where the classical collaborative filtering, and content-based filtering methods try to learn a static recommendation model given training data.

2.0 Content Block

UCB-ALP algorithm: The framework of UCB-ALP is shown in the right figure. UCB-ALP is a simple algorithm that combines the UCB method with an Adaptive Linear Programming (ALP) algorithm, and can be easily deployed in practical systems. It is the first work that show how to achieve logarithmic regret in constrained contextual bandits. Although is devoted to a special case with single budget constraint and fixed cost, the results shed light on the design and analysis of algorithms for more general CCB problems.

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Non-stationary bandit This framework refers to the multi-armed bandit problem in a non-stationary setting (i.e., in presence of concept drift). In the non-stationary setting, it is assumed that the expected reward for an arm k can change at every time step $t \in T$: $\mu_{t-1}^k \neq \mu_t^k$. Thus, μ_t^k no longer represents the whole sequence of expected (stationary) rewards for arm k . Instead, μ^k denotes the sequence of expected rewards for arm k , defined as $\mu^k = \{\mu_t^k\}_{t=1}^T$. A dynamic oracle represents the optimal policy to be compared with other policies in the non-stationary setting. The dynamic oracle optimises the expected reward at each step $t \in T$ by always selecting the best arm, with expected reward of μ_t^* . Thus, the cumulative expected reward $D(T)$ for the dynamic oracle at final time step T is defined as:

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Adversarial bandit Another variant of the multi-armed bandit problem is called the adversarial bandit, first introduced by Auer and Cesa-Bianchi (1998). In this variant, at each iteration, an agent chooses an arm and an adversary simultaneously chooses the payoff structure for each arm. This is one of the strongest generalizations of the bandit problem as it removes all assumptions of the distribution and a solution to the adversarial bandit problem is a generalized solution to the more specific bandit problems.

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$\rho = T \mu^* - \sum_{t=1}^T r_t$, where μ^* is the maximal reward mean, $\mu^* = \max_k \{\mu^k\}$, and r_t is the reward in round t . A zero-regret strategy is a strategy whose average regret per round ρ / T tends to zero with probability 1 when the number of played rounds tends to infinity. Intuitively, zero-regret strategies are

guaranteed to converge to a (not necessarily unique) optimal strategy if enough rounds are played.

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Constrained contextual bandit In practice, there is usually a cost associated with the resource consumed by each action and the total cost is limited by a budget in many applications such as crowdsourcing and clinical trials. Constrained contextual bandit (CCB) is such a model that considers both the time and budget constraints in a multi-armed bandit setting. A. Badanidiyuru et al. first studied contextual bandits with budget constraints, also referred to as Resourceful Contextual Bandits, and show that a $O(\sqrt{T})$ regret is achievable. However, their work focuses on a finite set of policies, and the algorithm is computationally inefficient.

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Online non-linear bandits UCBogram algorithm: The nonlinear reward functions are estimated using a piecewise constant estimator called a regressogram in nonparametric regression. Then, UCB is employed on each constant piece. Successive refinements of the partition of the context space are scheduled or chosen adaptively. Generalized linear algorithms: The reward distribution follows a generalized linear model, an extension to linear bandits. KernelUCB algorithm: a kernelized non-linear version of LinUCB, with efficient implementation and finite-time analysis. Bandit Forest algorithm: a random forest is built and analyzed w.r.t the random forest built knowing the joint distribution of contexts and rewards. Oracle-based algorithm: The algorithm reduces the contextual bandit problem into a series of supervised learning problem, and does not rely on typical realizability assumption on the reward function.

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There are two predominant settings in BAI: Fixed budget setting: Given a time horizon $T \geq 1$, the objective is to identify the arm with the highest expected reward $a^* \in \arg \max_k \mu_k$ minimizing probability of error δ . Fixed confidence setting: Given a confidence level $\delta \in (0, 1)$, the objective is to identify the arm with the highest expected reward $a^* \in \arg \max_k \mu_k$ with the least possible amount of trials and with probability of error $P(\hat{a}_T \neq a^*) \leq \delta$. For example using a decision rule, we could use m_1 where m is the machine no.1 (you can use a different variable respectively) and 1 is the amount for each time an attempt is made at pulling the lever, where $\sum_{i=1}^M m_i = M$, identify M as the sum of each attempts $m_1 + m_2 + \dots$ as needed, and from there you can get a ratio, sum or mean as quantitative probability and sample your formulation for each slots. You can also do $\sum_{k=1}^N \mu_k \propto \sum_{i=1}^N \mu_i - (n_j)$ where $m_1 + m_2$ equal to each a unique machine slot, x, y is the amount each time the lever is triggered, N is the sum of $(m_1 x, y) + (m_2 x, y) + \dots$, k would be the total available amount in your possession, k is relative to N where $N = n(a, b), (n_1 a, b), (n_2 a, b)$ reduced n_j as the sum of each gain or loss from a, b (for example, suppose you have 100\$ that is defined as n , and a would be a gain, b is equal to a loss. From there you get your results either positive or negative to add for N with your own specific rule) and i as the maximum you are willing to spend. It is possible to express this construction using a combination of multiple algebraic formulation, as mentioned above where you can limit with T for, or in time and so on.